Normalisation Recap

Normalisation, also known as **standardisation**, is a way of applying th
have measured two features for each data point, but one varies betwee
start adding these two together and making calculations, you can see th
in the second ones. This is very obvious when we think about proportio
feature, and 1 for the second one. If suddenly the first one were to be h
would still be 1. Conversely, if the second variable is halved, it is only go
900). The second one is so small that its changes won't really be visible
that, you transform them so that they are measured *on the same scale*,
0 and 1, shrinking or increasing your variables, but respecting all propor
and the second 0.5). In this case, halving one variable (to 0.45) or the ot
result!

The method is the following: you **subtract the mean**, and **divide by th**

$$X_{ij}^{Z} = \frac{x_{ij} - \bar{x}_j}{s_j}.$$

This will transform your data so that it now has a **mean of 0** and **a vari**
more about them here), and make sure that the *impact* is the same for
very large things, and then very small things, and want to work with all

If on the contrary you would rather have them distributed between 0 a
be used:

$$X_{ij}^{U} = \frac{x_{ij} - \min\limits_i x_{ij}}{\max\limits_i x_{ij} - \min\limits_i x_{ij}}.$$

Very often in Data Science, and especially using the K-means algorithm,
(for instance, one feature might be the age of a person, and another the
trying to make calculations on such datasets, it is important to think abo
lot, as ages will be greater numbers than the number of daily cups), and
normalisation before using other algorithms (after which, for instance,
preserving proportions, and allowing easier calculation between the two

Examples

Normalisation using one method or another is widely used in Data Scie
where min max scaling is used, taken from here:

- "*k-nearest neighbors with an Euclidean distance measure if want*
- *k-means (see k-nearest neighbors)*
- *logistic regression, SVMs, perceptrons, neural networks etc. if you otherwise some weights will update much faster than others*
- *linear discriminant analysis, principal component analysis, kernel*